



# MERN STACK INTERN

**Name:** Muneeb Ur Rehman

**Task Report:** (Advanced CSS)

**Project:** Responsive Feature/Team Showcase

**Module:** Week 2, Day 1 – Advanced CSS

**Focus:** Transforms, Transitions, and Keyframe Animations

## 1. Objective

To design and develop a responsive "Team Cards" section for a tech startup (DevPreneurs). The task required implementing visual polish through smooth hover interactions and page-load animations while maintaining optimal browser performance.

## 2. Technical Concepts Applied

- **Responsive Layout (CSS Grid):** Utilized `display: grid` with `auto-fit` and `minmax` to create a fluid layout that automatically adapts to mobile, tablet, and desktop screens without requiring complex media queries.
- **CSS Transitions:** Applied a 0.3s easing function to the cards. This acts as a bridge between the default state and the hover state, ensuring visual changes happen smoothly rather than instantly.
- **CSS Transforms:** Utilized 2D transforms (`scale` and `translateY`) to manipulate the physical space the cards occupy when interacted with, creating a tactile, "lifting" effect.
- **Keyframe Animations:** Created a custom `@keyframes` sequence to dictate how the elements behave the moment the DOM (Document Object Model) loads, moving them from an invisible, lowered state to their final resting position.

## 3. Implementation Details

The project was divided into a clear separation of concerns (HTML for structure, CSS for styling and animation). The core interactive features were implemented as follows:

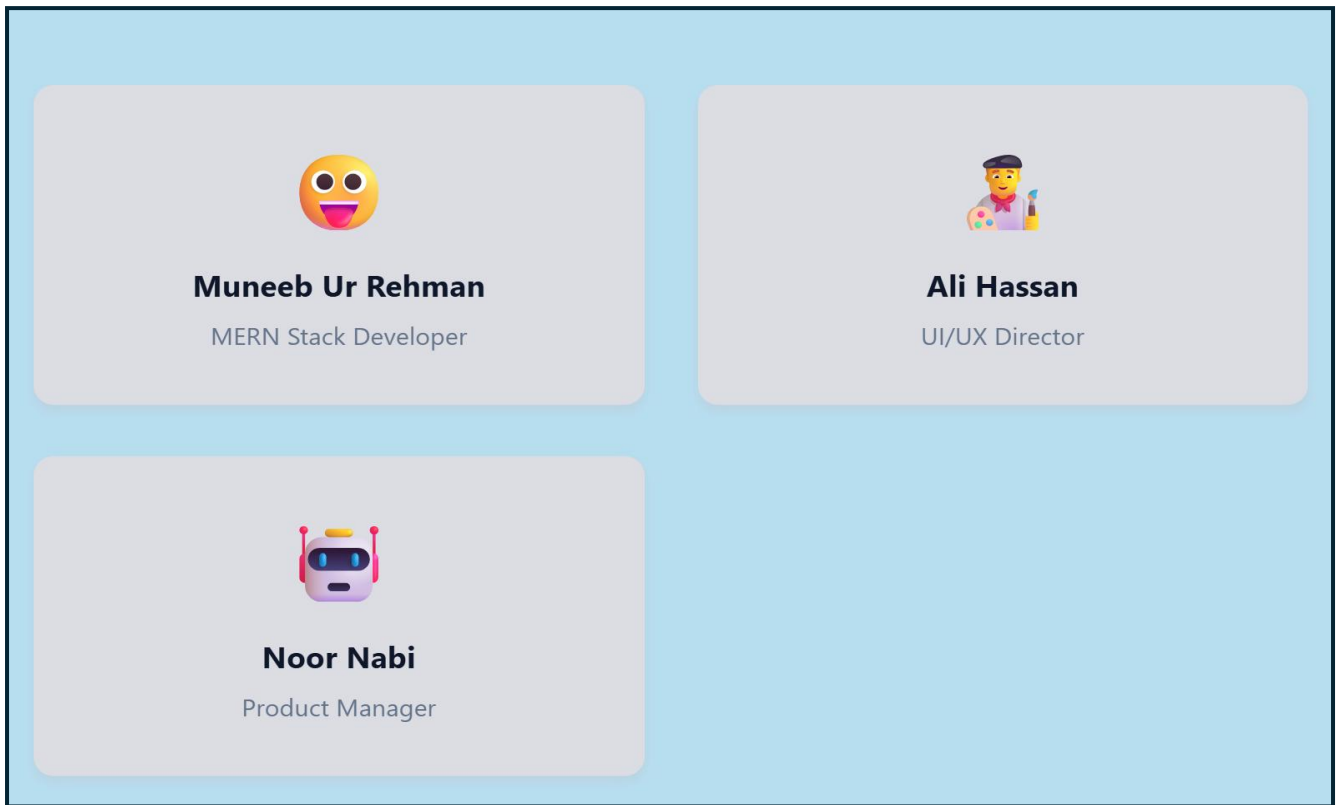
- **The Hover State:** Triggered via **THE .CARD:HOVER PSEUDO-CLASS**.
  - Applied transform: **`SCALE(1.05) TRANSLATEY(-5PX)`**; to enlarge the card by 5% and nudge it upwards by 5 pixels.
  - Increased the box-shadow depth to enhance the 3D lifting illusion.
- **The Entry Animation:** Created a custom animation named `fadeAndSlide`.
  - **0% (Start):** Card opacity set to 0 (invisible) and translated 30px down along the Y-axis.
  - **100% (End):** Card opacity set to 1 (fully visible) and translated to 0px (default position).
  - Applied the animation to the `.card` class with a 0.6s duration and a forwards fill mode so the cards remain visible after the animation finishes.

## 4. Performance Considerations

To ensure the visual polish did not sacrifice website performance, all animations and transitions were strictly limited to the `transform` and `opacity` properties. Because modern web

browsers can offload these specific properties to the device's GPU (Graphics Processing Unit), the animations run at a smooth 60 frames per second without causing layout recalculations or lag.

## 5. Output



## 6. GitHub Code Link

<https://github.com/955muneeb/Glaxit-internship/tree/main/week-2/day-1>